Meta Data

Table 1: original: crop_yield.csv

#	Column Name	Type	Description
1	Region	object	Geographical region
2	Soil_Type	object	Type of soil
3	Crop	object	Type of crop
4	Rainfall_mm	float64	Annual rainfall (mm)
5	Temperature_Celsius	float64	Average temp (°C)
6	Fertilizer_Used	bool	Fertilizer indicator
7	Irrigation_Used	bool	Irrigation indicator
8	Weather_Condition	object	Predominant weather
9	Days_to_Harvest	int64	Days to harvest
10	Yield_tons_per_hectare	float64	Yield (TARGET)

Table 2: Original Dataset - agriculture_farming.csv

#	Column Name	Type	Description
1	Farm_ID	object	Unique identifier
2	Crop_Type	object	Crop type
3	Farm_Area(acres)	float64	Total area
4	Irrigation_Type	object	Irrigation method
5	Fertilizer_Used(tons)	float64	Total fertilizer
6	Pesticide_Used(kg)	float64	Total pesticide
7	Yield(tons)	float64	Total yield
8	Soil_Type	object	Soil type
9	Season	object	Growing season
10	Water_Usage(m3)	float64	Total water usage

Table 3: Predicitive_final_dataset (merged and derived columns)

#	Column Name	Type	Source
1	Region	object	Merged
2	Soil_Type	object	Merged
3	Crop	object	Merged
4	Rainfall_mm	float64	Merged
5	Temperature_Celsius	float64	Merged
6	Weather_Condition	object	Merged
7	Days_to_Harvest	int64	Merged
8	Yield_tons_per_hectare	float64	Merged (Target)
9	Farm_Area(acres)	float64	Merged
10	Fertilizer_Used(tons)	float64	Merged
11	Pesticide_Used(kg)	float64	Merged
12	Water_Usage(cubic meters)	float64	Merged
13	Fertilizer_per_acre	float64	Feature Engineered
14	Pesticide_per_acre	float64	Feature Engineered
15	Water_per_acre	float64	Feature Engineered
16	Total_Water_Available	float64	Feature Engineered
17	Moisture_Heat_Index	float64	Feature Engineered
18	Growing_Degree_Days	float64	Feature Engineered
19	Input_Cost_Index	float64	Feature Engineered
20	Heat_Stress_Flag	int32	Feature Engineered
21	Drought_Risk_Flag	int32	Feature Engineered
22	Weather_Risk_Score	int32	Feature Engineered

Table 4: Derived Features: Formulas, and Purpose

#	Feature Name	Category	Formula	Purpose
1	Fertilizer_per_acre	Efficiency	Fertilizer_Used(tons) / Farm_Area(acres)	Input intensity normalization
2	Pesticide_per_acre	Efficiency	Pesticide_Used(kg) / Farm_Area(acres)	Input intensity normalization
3	Water_per_acre	Efficiency	Water_Usage(cubic meters) / Farm_Area(acres)	Input intensity normalization
4	Total_Water_Available	Climate	Rainfall_mm + Water_Usage(cubic meters)	Total moisture availability
5	Moisture_Heat_Index	Climate	Rainfall_mm / (Temperature_Celsius + 1)	Water–temperature interaction
6	Growing_Degree_Days	Climate	Temperature_Celsius × Days_to_Harvest	Cumulative growth potential
7	Input_Cost_Index	Composite	Fertilizer_per_acre + Pesticide_per_acre + Water_per_acre	Combined resource intensity
8	Heat_Stress_Flag	Risk	Temperature_Celsius > Q0.75(Temperature_Celsius)	High temperature stress indicator
9	Drought_Risk_Flag	Risk	Rainfall_mm < Q0.50(Rainfall_mm)	Low rainfall indicator
10	Weather_Risk_Score	Risk	Heat_Stress_Flag + Drought_Risk_Flag	Combined environmental risk